

Switching Circuits Laboratory (CS29002)

Assignment 7 (18-03-2026)

- a) Design a 4-bit bidirectional shift register using D flip-flops. Configure the register in right-shift mode, apply the pattern 00110011... in synchronism with a slow clock, and observe the outputs on the on-board LEDs to verify operation.
- b) Modify a 4-bit right-shift register to implement:
- i) A 4-bit ring counter. Initialize the register with 1000, apply a clock signal of sufficiently high frequency, and trace the outputs by observing on the oscilloscope.
 - ii) A 4-bit linear feedback shift register that realizes the following characteristic polynomials:
 - $f(x) = x^4 + x + 1$
 - $f(x) = x^4 + x^3 + x + 1$

Initialize the LFSR suitably, and observe the 4-bit patterns that are generated on on-board LEDs by applying a slow clock.

Instructions:

- 1) *Draw the circuit diagram on paper, following the suggested conventions. Following this, you can issue the components and start with the realization on the breadboard.*
- 2) *Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group. **There must not be any kind of plagiarism in the preparation of the report.***